Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

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Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

A drafter program that helps students perform the Euler cromer method in steps to calculate velocity, position, time and momentum close to light speed. It will take the user to a page that explains why the velocity is leveling off at speed of light. It can also use a loop for multiple time steps to show the leveling off.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Students could integrate or use desmos but this tool would help them learn the steps of the method they would need to perform in physics class and on the exam.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

This application would be used to teach students how to do the method and can be used infinitely many times as the student would input the values to use practice problems. Thai could help students remember the steps or check their calculations as they practice.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The main goal is to use loops and pages as steps in teaching students how to perform the euler cromer method, allowing learning students to check answers and see calculations for many steps to see what happens as a particle reaches the speed of light.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will input the specific information from their practice problem provided by a teacher or professor or values from a practice problem they personally created.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It will perform the euler cromer method. This method takes in net force and time step value to commute momentum, velocity and displacement. It will also put the data it commutes into a list and make a graph.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will receive a graph, and each step will have a value the user should have commuted themself to check their process.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Momentum, Velocity, Time, Displacement, Gamma(used in solving for momentum)

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Momentum: 1. *Net Force* 2. *Current speed* 3. *Gamma* 4. *Initial Momentum*

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

I think it will be the most difficult to create the graph throughout all the steps and show it at the end.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I plan to use a mix of generated code and coding it myself. I plan to work on Monday of that week from 3-5, Tuesday from 11-12 and if needed from 2-3. To avoid these hallucinations, I plan to verbally explain to myself what each part does individually and keep a log of what I generated to remember what might need extra reinforcement.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

I want to understand what the different functions I can use in Drafter are. I do not currently have a great understanding of Drafter so this project should help me understand how to use it.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a tool to teach students how to use the euler cromer method to teach students the method and model the change in displacement and horizontal asymptote at the speed of light. What are some features I could include to help the learning process?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

A graph that models the asymptote at the speed of light that is dependent on the information the student puts in. A simulation that models the particle moving.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

The simulation would be very difficult to do because it would include an animation that I have not learned to do yet. Would just the graph be a good visual or do you think more would be needed.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

It has made me feel more focused on using data from the user to model a particle in a graph while enforcing the mathematical method. I liked the simulation idea but I think that is too advanced for my capabilities in drafter.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will talk to the TAs or will talk to an LLM to figure out why my program is not working without it telling me exactly how to fix it.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I plan to first attempt to trace my program by myself. I will look at all of the drafter resources to figure out if there is an example close to my issue. I can talk to other people in my class and find out if they can guide me with solving my problem without telling me the answer. I can always go to the TA Office or Professor office hours for additional help.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use an LLM

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.